

ALEXANDRIA INTL, Egypt

WMO: 623180

Lat: 31.184N

Lon: 29.949E

Elev: -6

StdP: 14.70

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.7	46.6	35.4	30.3	56.6	37.7	33.1	56.5	28.2	56.2	25.0	56.3	3.3	190	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	93.0	71.7	89.7	73.8	88.0	74.4	78.2	86.5	77.2	85.6	76.4	84.7	9.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	133.4	83.6	74.8	130.1	83.2	73.6	124.9	82.5	41.7	86.3	40.7	85.6	39.9	84.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	19.0	17.1	40.3	103.1	3.0	4.1	38.2	106.1	36.5	108.4	34.8	110.7	32.7	113.7		
(5)			38.4	80.4	2.5	1.5	36.6	81.5	35.1	82.4	33.7	83.2	31.9	84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.9	57.2	58.1	61.7	66.7	72.3	77.7	81.1	82.0	79.6	74.3	67.1	60.3
	DBStd	9.48	3.12	3.79	4.28	4.89	4.37	3.19	2.07	1.99	2.74	3.50	4.01	3.67
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	761	242	195	120	31	1	0	0	0	0	0	21	151
	CDD50	7270	224	228	364	500	692	832	964	991	887	755	513	320
	CDD65	2556	0	3	19	81	228	382	499	526	437	290	85	6
	CDH74	22982	6	35	172	568	1363	3063	5212	5997	4349	1894	307	16
	CDH80	6964	0	5	53	209	443	840	1652	2085	1281	356	38	2
(14) Wind	WSAvg	8.8	8.0	8.5	9.0	9.3	9.3	9.8	10.4	9.7	8.8	8.0	7.2	7.4
(15) Precipitation	PrecAvg	6.3	1.9	1.1	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.3	1.1	1.4
	PrecMax	13.2	4.4	3.0	1.5	0.4	0.2	0.0	0.0	0.0	0.3	1.9	4.5	4.5
	PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.7	1.3	0.9	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.4	1.3	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	78.9	86.2	94.6	98.7	96.8	94.6	92.9	95.2	90.9	84.3	75.6
		MCWB	58.2	60.7	64.5	67.2	69.0	71.2	74.2	75.9	73.4	70.0	67.4	63.6
	2%	DB	68.4	72.3	78.5	85.6	89.6	89.8	89.7	89.8	89.9	86.0	79.7	71.9
		MCWB	57.9	58.6	61.6	65.6	67.3	71.6	75.7	76.5	73.7	71.0	67.8	60.8
	5%	DB	66.5	68.3	73.3	80.4	84.3	86.3	87.9	88.2	87.6	83.6	77.1	70.0
		MCWB	57.2	57.8	60.7	63.6	67.6	72.5	75.4	76.2	74.2	71.0	66.1	60.3
	10%	DB	64.7	66.1	69.7	75.4	80.5	84.3	86.3	87.6	85.8	81.6	75.2	68.2
		MCWB	56.1	56.8	59.6	62.8	67.2	72.0	74.7	76.0	73.5	70.0	64.8	59.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.2	64.2	66.9	69.6	72.4	76.9	79.0	79.7	78.1	75.6	72.1	66.0
		MCDB	68.2	74.6	80.5	85.8	85.3	86.5	88.0	87.4	86.8	84.1	78.7	72.5
	2%	WB	59.6	60.8	63.6	67.1	70.6	75.2	77.4	78.7	76.7	73.8	69.6	63.0
		MCDB	66.5	68.9	74.6	81.4	81.9	84.5	86.2	86.7	85.1	81.9	76.5	69.6
	5%	WB	58.5	59.2	61.7	65.1	69.3	74.2	76.6	77.7	75.6	72.4	67.8	61.2
		MCDB	65.2	66.3	70.5	76.4	79.8	83.1	85.2	85.8	84.1	80.7	74.7	68.1
	10%	WB	57.2	57.7	60.2	63.5	68.2	73.1	75.6	76.9	74.7	71.2	66.1	60.0
		MCDB	63.6	64.6	68.0	73.3	78.1	81.7	83.9	85.0	83.3	79.7	73.7	66.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.1	16.2	17.1	15.9	13.0	11.3	11.4	13.3	14.7	15.7	15.8
		MCDBR	19.0	21.5	24.5	27.7	25.2	17.4	13.5	12.7	17.3	18.3	18.6	18.2
		MCWBR	11.0	11.3	11.3	11.1	9.3	6.9	5.6	5.3	7.0	7.8	9.2	10.3
	5% WB	MCDBR	17.6	18.7	20.5	23.2	18.9	14.3	12.4	12.0	14.7	16.6	17.2	16.6
		MCWBR	11.1	10.7	10.5	10.3	8.1	6.4	5.6	5.6	6.7	8.0	9.1	10.2
(40) Clear-Sky Solar Irradiance	taub		0.393	0.434	0.461	0.506	0.499	0.433	0.422	0.430	0.437	0.431	0.401	0.391
	taud		2.234	2.094	2.022	1.927	1.983	2.224	2.264	2.247	2.225	2.227	2.290	2.269
	Ebn at Noon		258	258	260	253	255	272	274	271	265	256	254	252
	Edh at Noon		37	46	53	61	58	45	43	43	43	40	35	34
(44) All-Sky Solar Radiation	RadAvg		1014	1284	1726	2098	2357	2530	2469	2255	1947	1499	1117	920
	RadStd		46	80	60	72	67	37	24	25	26	49	52	37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.83	N/A	N/A	+1.12	+0.77	N/A	N/A	-122	+347	+214
(47) Regional (4 neighbors)	+0.99	N/A	-1.17	+0.95	+0.67	N/A	N/A	-104	+358	+254

Nomenclature: See separate page